SISB4301 DESIGN THINKING AND INNOVATIONS

AUTONOMOUS LINE FOLLOWING ROBOT WITH OBSTACLE DETECTION AND BLOWER MECHANISM

Submitted in partial fulfilment of the requirements for the award of Bachelor of Engineering degree in Electronics and Communication Engineering

by

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

SCHOOL OF ELECTRICAL AND ELECTRONICS

SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY

(DEEMED TO BE UNIVERSITY)

CATEGORY -1 UNIVERSITY BY UGC

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JEPPIAAR NAGAR, RAJIV GANDHI SALAI, CHENNAI - 600 119

November 2025 to April 2026

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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the bonafide work of **NITHIN VINAYAK G K [44130318], NITHISHKUMAR K [44130323], PALURI NIKHIL [44130331]** and **PUNITHUKUMAR A [44130360]** who carried out the project entitled "**AUTONOMOUS LINE FOLLOWING ROBOT WITH OBSTACLE DETECTION AND BLOWER MECHANISM**" under my supervision from November 2025 to April 2026.

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DECLARATION

We, **NITHIN VINAYAK G K [44130318], NITHISHKUMAR K [44130323], PALURI NIKHIL [44130331] and PUNITHUKUMAR A [44130360]** hereby declare that the Project Report entitled **AUTONOMOUS LINE FOLLOWING ROBOT WITH OBSTACLE DETECTION AND BLOWER MECHANISM** is done under the guidance of **DR. N. R. KRISHNAMOORTHY, M.E., Ph.D.,** is submitted in partial fulfilment of the requirements for the award of Bachelor of Engineering degree in **Electronics and Communication Engineering.**

DATE: 25/04/2026

PLACE: Chennai

ACKNOWLEDEMENT

I am pleased to acknowledge my sincere thanks to Board of Management of **SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY** for their kind encouragement in doing this project and for completing it successfully. I am grateful to them.

I convey my profound gratitude to **Dr. N. M. NANDHITHA, M.E., Ph.D.,** Prof. & Dean, School of Electrical and Electronics and **Dr. T. RAVI, M.E., Ph.D.,** Head of the Department of **Electronics and Communication Engineering**, Sathyabama University for having been a constant source of support and encouragement for the completion of the project.

I consider my self-most fortunate and immensely privileged to have worked under the guidance of **Dr. N. R. KRISHNAMOORTHY, M.E., Ph.D.,** Department of **Electronics and Communication Engineering** to complete this successfully.

I wish to express my thanks to all Teaching and Non-teaching staff members of the Department of Electronics and Communication Engineering who were helpful in many ways for the completion of the project.

ABSTRACT

This project presents an autonomous mobile robot designed to patrol predefined paths, detect fires, extinguish small flames, and provide real-time updates to a cloud-based monitoring platform. The primary objective is to enhance safety and minimize damage in high-risk environments such as warehouses, factories, and storage facilities, where fire incidents can lead to significant property loss and pose serious threats to human lives.

The robot utilizes a 5-channel Infrared (IR) sensor array for accurate line following, enabling it to navigate reliably along a designated path. At its core is an ESP32 microcontroller, which serves as the central processing unit. It continuously processes inputs from the line-following sensors and flame detection sensors, and controls the robot's movement through an L298N motor driver.

Upon detecting a flame, the robot automatically switches from navigation mode to fire-fighting mode. A DC blower, switched via a MOSFET, is activated to direct a stream of air at the flame, effectively extinguishing small fires. The built-in Wi-Fi capability of the ESP32 allows the system to transmit real-time data—including fire alerts, robot location, and battery status—to cloud platforms such as ThingSpeak, Blynk, or similar IoT services. This enables remote monitoring, instant notifications, and comprehensive data logging for analysis and record-keeping.

The entire system is powered by a Li-ion battery, with a buck converter ensuring stable and efficient voltage regulation for all components.

By integrating embedded systems, sensor technology, motor control, wireless communication, and IoT, this project offers a compact, intelligent, and scalable solution for proactive fire monitoring and response. It demonstrates how autonomous robotics can significantly improve safety and operational efficiency in industrial and commercial settings through real-time remote supervision.

***KEYWORDS*:** Autonomous Line Following Robot, Flame Sensor, IR Sensor, ESP32 Microcontroller, ThingSpeak, Blynk, Li-ion Battery, DC Blower, GitHub READMEs, MOSFET, L298N Motor, Buck Converter, IOT;

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	
1.	INTRODUCTION 1.1.Problem Statement 1.2. Objective 1.3. Motivation	
2.	LITERATURE SURVEY	
3.	PROPOSED METHODOLOGY 3.1. Block Diagram 3.2. Hardware 3.3. Software	
4.	IMPLEMENTATION 4.1. System Design Overview 4.2. Hardware Integration 4.3. Software Design 4.4. System Flow 4.5. Testing & Calibration 4.6. Final Implementation	
5.	RESULTS & DISCUSSION 5.1. Advantages 5.2. Limitations 5.3. Future Scope	
6.	APPLICATIONS	
7.	CONCLUSION	
8.	REFERENCES	

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
3.1.	Block Diagram	
3.2.	ESP32 Controller	
3.3.	IR Sensor	
3.4.	IR Flame Sensor	
3.5.	DC Gear Motor	
3.6.	DC Blower	
3.7.	Motor Driver Module	
3.8.	Li-ion Battery	
3.9.	Buck Converter	
3.10.	Car Chassis	
3.11.	Relay Module	
3.12.	Charger Module	
3.13	Arudino IDE	

1. INTRODUCTION

In the role of scientific development taking place today in this world, robots have become one of the basic requirements for use in various fields of industry, but that the use of the robot was not limited to industry alone, but also entered in the field of restaurants, laboratories, hospitals, educational fields, etc. because the robot works with very high accuracy and reduces human interference and human errors in a way Large and of course in the recent period witnessed great competition between companies and restaurants to distinguish diversity and development in order to attract consumers, whether in terms of providing service or quality. Using the security can reduce the operating costs that the project owner incurs. And that is by creating a robot that makes two or three people to the fullest and with very high precision. These robots cannot be tired or exhausted, nor are they affected by human emotions. The work of these robots is through a sensor that follows a line drawn on the ground of the place. This line can be straight or oblique. This robot can perform its tasks in the most difficult and worst conditions without feeling any fatigue. These are usually 2-wheeled robots, a sensor, and a left sensor. The sensor (ground sensor) is a device that explores and obeys some types of orders from the environment around it. These orders can be like (temperature, humidity or other environmental factors) and the output is by means of a signal that is sent to the screen and the user or the beneficiary of this process can read it very clearly through the sensor site or it is sent electronically through the network. Today, all known sensors convert the measurement of environmental and physical factors into electrical ones. Of course, mechanically, the robot consists of two rear wheels rotated by a process that runs on a continuous current, but the front wheel is free movement and depends in its movement and push it on the strength of the two rear wheels. Also, it contains an Arduino board and a sensor that senses the line on the floor and many continuous attempts are made on the robot to achieve the highest ratio on following the line on the ground from the starting point (A) to the access point (B).

Problem Statement:

Many industries face the following challenges:

- Manual fire suppression exposes workers to dangerous conditions.
- Fixed fire systems cover only limited areas.
- Emergency response time may be delayed.
- Fire detection often requires human monitoring.
- Traditional fire systems involve high installation and maintenance cost.

There is a need for a mobile, autonomous system that can continuously monitor specific areas, detect fire early, and respond immediately. This system should reduce human exposure, speed up response times, and function reliably in controlled industrial environments.

OBJECTIVES:

- To make robot move by following a line using IR Sensors.
- To detect fir and know its direction using flame Sensors.
- To make robot turn towards fire automatically and blow air to off the fire.
- To control robot using a step-by-step program in ESP32.

Primary Objectives:

- Design a robot that follows aa predefined line path autonomously.
- Integrate flame sensors to detect fire with directional awareness.
- Develop an automatic fire suppression mechanism using DC Blower.
- Implement real time control logic to switch between navigation and firefighting models.

Secondary Objectives:

- Reduce power consumption for longer operations.
- Improve system reliability under different indoor conditions.
- Design a system architecture that supports future expansions.
- Test and evaluate system performance under controlled conditions.

Scope:

- Designed for indoor environments such as warehouses and factories.
- Suitable for detecting and extinguishing small scale fires.
- Operates on predefined line based navigation.
- Developed as a single autonomous unit with potential for scaling.

Motivation:

In time of automation advances to reduce human efforts, it is necessary to develop colour line following robot, this robot can be used in airports to carry equipment and baggage from one place to another place, and it can be used for home automation. Advantage of such robots is that they can operate efficiently for 6-7 long hours with a single charge. Thus, it is profitable for the business itself. Therefore, for a large country like India, it is necessary to use the line following robots in restaurants, industries etc. Such Robots come into play when large and heavy machineries are to be transferred from one place to another within industries. This technology can be implemented in running buses or other mass transit systems.

A Robot is a machine which is completely automatic, i.e. it starts on its own, decides its own way of work and stops on its own. Robotics has greatly advanced in the developed countries. High performance, high accuracy, lower labour cost and the ability to work in hazardous places have put robotics in an advantageous position over many other such technologies but as for developing countries like Bangladesh it is still quite out of reach. But it is one of the most fascinating and interesting aspects to the new generations and a lot of development in robotics has been done in last couple of years. Robots have several useful applications in our daily. It is actually a replica of human 4 being, which has been designed to ease human burden. It can be controlled pneumatically or using hydraulic ways or using the simple electronic control ways.

2. LITERATURE SURVEY

Design of autonomous line following robot with obstacle detection and blower mechanism. This paper shows design and implementation of the Line Follower Robot and its ability to follow black line and possess the ability to detect the presence of an obstacle and choose the other path and the ability to detect the presence of fire and blow it. It's programmed in such a way that instructions are given to the robot which senses the line and attempts to move towards the target. The robot can easily move along very congested curves as it continuous its data from the sensors. This robot avoids collision and it can detect collision with an obstacle sensor and blow out fire if it comes into its path.

This paper outlines the development and application of a line following robot with the blower in a healthcare management system. The robot is designed to transport medical supplies and equipment in hospitals and clinics, and is capable of navigating through narrow corridors and tight spaces, and is capable to extinguish small fires in narrow areas of the management.

The robot is designed to be small and lightweight, and is able to navigate through narrow spaces with ease. The paper provides details on the hardware and software components used in the robot, as well as the algorithm used for line following. This paper reviews various image-based obstacle detection and blowing of fire techniques used in unmanned vehicles. The reviewed techniques are divided into monocular and stereo methods based on whether a single or multiple synchronized sensors are used for obstacle detection and blowing. The paper provides detailed descriptions of each technique, including their advantages and limitations. The experimental results of each technique are also provided to validate their effectiveness in those techniques.

Overall, these research papers provide valuable insights into the design and implementation of line follower robots with advanced capabilities for obstacle detection and avoidance.

3. PROPOSED METHODOLOGY

3.1. BLOCK DIAGRAM:

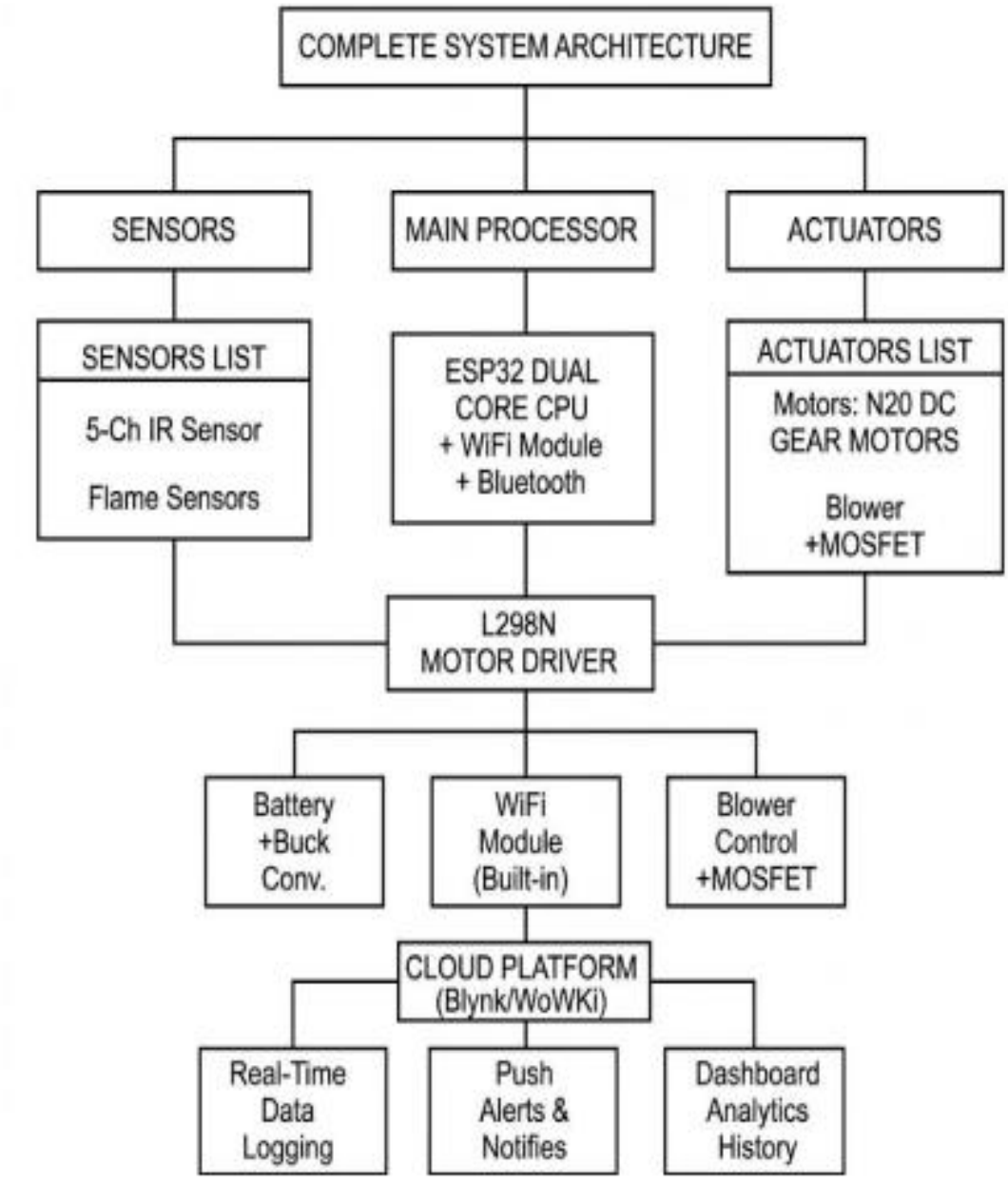


Fig. 3.1. Block Diagram

Now, we'll talk about 2 main parts. The 1st part talks about the parts of the robot (also known as Hardware). As for the 2nd part, it takes about Arduino IDE program (also known as Software).

3.2. HARDWARE:

The components consist are generalised as Controller, Sensors, Actuators, Motor Control, Power System, Mechanical Parts and Electronics.

1. ESP32 Controller: -

ESP32 is a family of low-cost, energy efficient microcontrollers that integrate both Wi-Fi and Bluetooth capabilities. These chips feature a variety of processing options, including the Tensilica Xtensa LX6 microprocessor available in both dual-core and single-core variants, the Xtensa LX7 dual-core processor, or RISC-V microprocessor. In addition, the ESP32 incorporates components essential for wireless data communication such as built-in antenna switches, an RF balun, power amplifiers, low-noise receivers, filters, and power-management modules. Typically, the ESP32 is embedded on device-specific printed circuit boards or offered as part of development kits that include a variety of GPIO pins and connectors, with configurations varying by model and manufacturer. The ESP32 was designed by Espressif Systems and is manufactured by TSMC using their 40 nm process. It is a successor to the ESP8266 microcontroller.

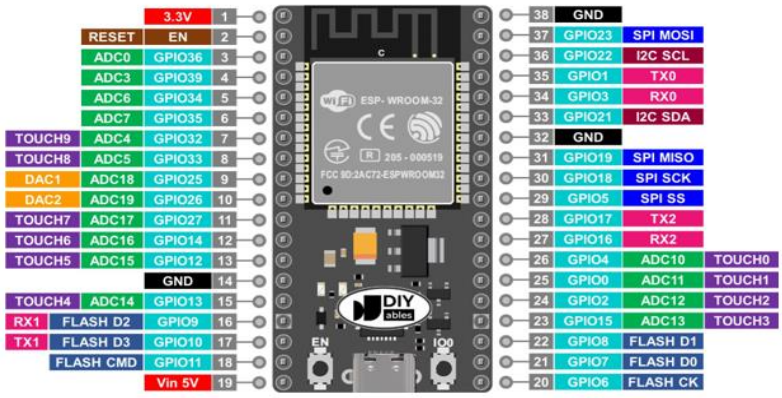


Fig. 3.2. ESP32 Controller

It exposes more pins than the 30-pin board (adds about 8 extra pins). Most useful pins are the standard GPIOs for sensors, motors, etc.

Warning: Out of the extra pins, GPIO6 to GPIO11 are internally connected to the flash memory. Do not use them as normal GPIOs — using them can brick (damage) your board or cause crashes.

It's a good choice as it has plenty of GPIOs for 5-channel IR sensor array + flame sensor. Built in Wi-Fi and enough pins left for motor driver and other accessories.

2. IR Line Sensor Array: -

This IR Sensor Array module is based on TCRT5000 sensors, making it a great detector for a line-following robot. It outputs a fast response time and high sensitivity for precise movement control. It easily integrates into DIY robotics, smart cars, and automated guidance systems.

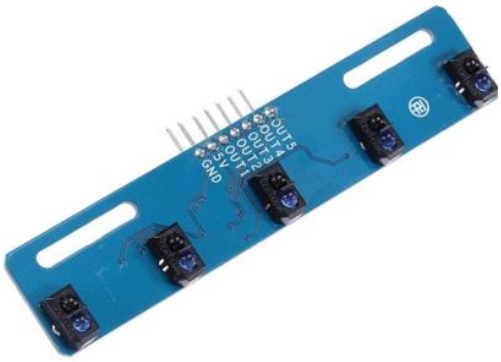


Fig. 3.3. IR Sensor

This TCRT5000 IR sensor module contains 5 IR LED/phototransistor pairs. Each sensor has a separate digital I/O measurable output. This reflection sensor array is designed to be used as a line sensor, but it may also be used as a proximity or reflection sensor. The module serves as a carrier for five IR emitters and receivers (photo-transistor) pairs. The infrared-emitting diode of the TRCT5000 sensor continuously emits infrared rays. When the emitted infrared rays are reflected by the object, they are received by the infrared receiver and output analogue values. The output analogue value is related to the object distance and the colour of the object. The position of the tracking line is judged by calculating the analogue value of the five outputs.

Here, 5-independent sensors, each operates at 3.3-5V. Output is Digital, adjusts to sensitivity, low power consumption and compact in size.

3. IR Flame Sensor: -

This flame sensor or fire sensor module works on the concept that when a flame or fire is burning it emits IR signals. This IR signal is then received by the IR receiver on the fire sensor module to detect the flame or fire. The sensor has an operating voltage from 3V to 5.5V and has both digital and analog output. The sensitivity of the digital output can be controlled by the on-board potentiometer. Detection angle of sensor is 60 degree and range is theoretically 100cm but practically you can get up to 20-30cm. This flame sensor consists of a photodiode, resistor SMD package of value 10K-2N0, capacitor SMD package of value 100nF – 2No, resistor SMD package of value 1K-2N0, potentiometer of value 10K- 1No, SMD led-2No.

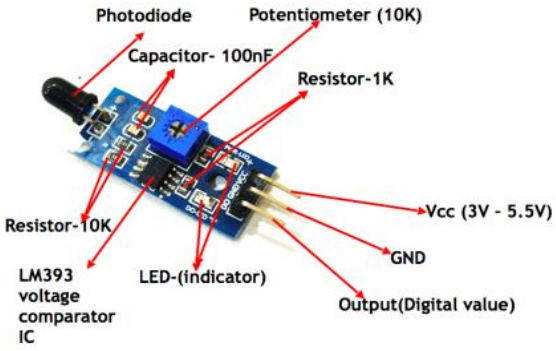


Fig. 3.4. IR Flame Sensor

Flames emit IR light mainly in the **760 nm to 1100 nm** wavelength range. The module uses a photodiode to detect this IR radiation. The signal goes through a comparator (op-amp), which compares it against a threshold you set with the on-board potentiometer. It detects fire at an angle of 60° and up to a distance of 80-100cm.

- Stronger flame → higher IR intensity → analog output drops.
- When IR exceeds the threshold → **digital output** goes **LOW** (fire detected).
- No flame → digital output stays **HIGH**.

It also responds to bright lights, sunlight, or remote controls, so it's best for short-range indoor or controlled use.

It’s commonly used in fire alarm systems, fire-fighting robots, safety monitoring and automatic extinguishers (in this robot). But some limitations are that it can be affected by sunlight, limited detection range, and only applicable for small flames. Here, 3 flame sensors are used so that it can able to detect flames in 3 directions – left, centre and right.

4. DC Gear Motor: -

A DC Gear motor is an electrical motor that uses direct current (DC) to produce mechanical force. The most common types rely on magnetic forces produced by currents in the coils. Nearly all types of DC motors have some internal mechanism, either electromechanical or electronic, to periodically change the direction of current in part of the motor.

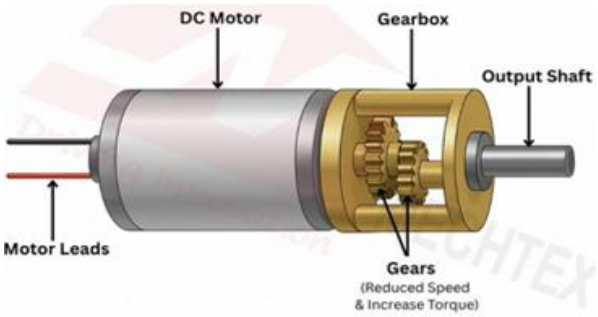


Fig. 3.5. DC Gear Motor

DC gear motor consists of a permanent magnet or electromagnet stator and a mature wound with the coil. When an electric current flows through the armature coils, it creates a magnetic field that interacts with the stator and generates torque that causes the armature to rotate. The gearbox attached to the DC motor contains a series of gears that convert the motor's high speed and low torque output to low speed and high torque output. These gears can be of various types such as spur gears**,** planetary gears, and helical gears, each offering different benefits in terms of efficiency, noise reduction, and load capacity. The main difference between a plain DC Motor and DC Gear Motor is that it produces lower output speed than plain motor but it has higher torque than plain motor.

Here, 2 DC Gear Motors are used with each has a voltage of 6-12V and has a RPM of 30-1000RPMs.

5. DC Blower: -



Fig. 3.6. DC Blower

A DC blower is a small electric device that uses a DC motor to produce a focused stream of air. Unlike a normal fan (which spreads air in all directions), a blower pushes air in a specific direction with higher pressure, making it ideal for targeted applications. When DC voltage is applied, the motor spins. This motor rotates in internal fan. Air is sucked in from one side and blown out forcefully through an outlet.

Here, it operates on DC voltage of range of 5-12V, works at a speed of 5000-8000RPMscompact and lightweight and provides high speed airflow. When the flame sensor detects fire, the ESP32 microcontroller turns ON the blower. It pushes air to put out the flame.

6. Motor Driver Module: -

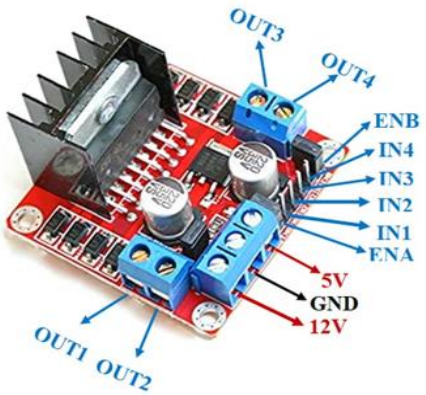


Fig. 3.7. Motor Driver Module

A **motor driver**, also known as a **control motor**, is an electronic device or module that controls and manages the operation of an electric motor. It serves as an interface between a microcontroller or other control system and the motor itself, enabling precise control of the motor's speed, direction, and other parameters. Motor drivers are commonly used in various applications, including robotics, automation, automotive systems, and industrial machinery. This module comprises two servo and two DC motor drivers. The motors are devoid of + and – terminals, thereby eliminating the need to be concerned about reverse connections. The direction of the motors changes based on the wiring.

The driver used here is L298N. Because it uses bipolar transistor, so it has relatively high voltage drop (~2V per bridge). Its supply voltage ranges from 5-35V, contains regulators, fly back diodes for inductive protection and power LED.

7. Li-ion Battery: -

A lithium ion battery is a type of rechargeable battery that uses reversible intercalation of Li^+ ions into electronically conducting solids to store energy. Compared to other types of rechargeable batteries, they generally have higher specific energy, energy density, and energy efficiency and a longer cycle life and calendar life. During charging, Li ions move from cathode to anode, and during discharging, Li ions move from anode to cathode and this movement generates electric current to power devices.

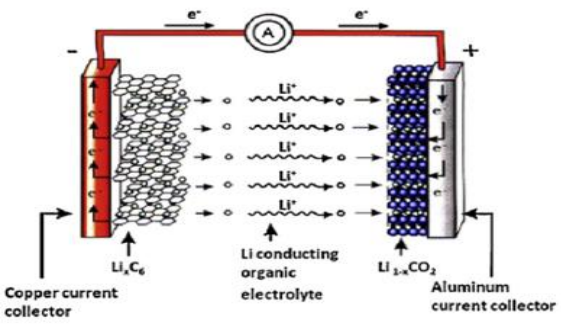


Fig. 3.8. Li-ion Battery

It's used here as it has high energy density, lightweight, long cycle life and no memory effect. A 7.4V battery is used which has an efficiency of some 80-90%.

8. Buck Converter: -

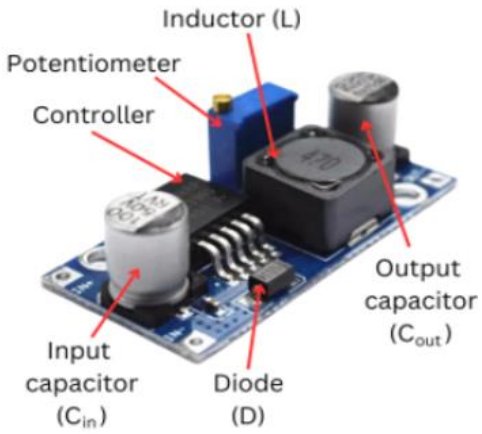


Fig. 3.9. Buck Converter

The buck converter, also referred to as a step-down converter, is a popular topology in power electronics that converts a higher input voltage to a lower output voltage. It is crucial in various applications, from portable devices to automotive systems, where specific components or subsystems require a lower voltage level to operate. The primary advantage of the buck converter is its simplicity, which enables efficient voltage conversion using a relatively small number of components. The operating principle of the buck converter involves controlled energy transfer from the input to the output through switches, an inductor, and a capacitor. A high-side switch (usually a MOSFET) and a low-side switch (typically a diode) are employed in the buck converter to control the current flow through the inductor. By adjusting the duty cycle of the high-side switch, the average output voltage can be regulated proportionally to the input voltage.

The module used here is LM2596. It helps to convert high DC voltage into lower DC voltage so that the microcontroller (12V → 5V) can work. It has high efficiency of 80-95%, less heat generation, compact size.

9. Car Chassis: -

An acrylic 2-wheel robot car chassis is a lightweight base platform used to build small mobile robots. It consists of 2 DC geared motors, 2 wheels, 1 caster wheel for balance and motor mounts. Acrylic is used as it's lightweight easy to drill & modify, cheap and enough space to place the sensors.



Fig. 3.10. Car Chassis

10. Relay Module & Resistors: -

A relay module is an electronic switching device that allows a low-power signal to control a high-power circuit. It works when a small voltage from Arduino is applied to the relay input. The relay coil gets energized. This creates a magnetic field that opens or closes the switch contacts which in turn as ON/OFF.

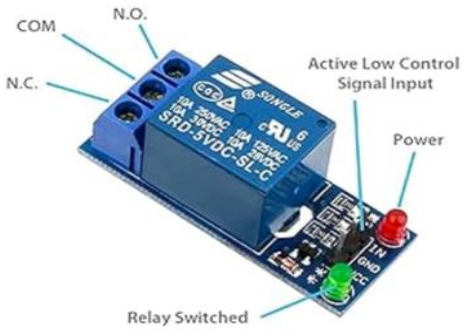


Fig. 3.11. Relay Module

Resistors are connected to the Relay Module.
It limits sudden current into module in which protects the microcontroller. Prevents random switching. Resistance Values varies from 220-330Ω.

11. Charger Module: -

A charger module is a compact circuit board designed to safely charge rechargeable batteries, especially single-cell Lithium ion batteries. It features a constant current or voltage charging, overcharge & over-discharge protection and can be connected with Micro-USB or Type-C input. It's low in cost byt it has a maximum limited current supply of 1A, only supports single-cell Li-ion battery and needs proper heat dissipation.

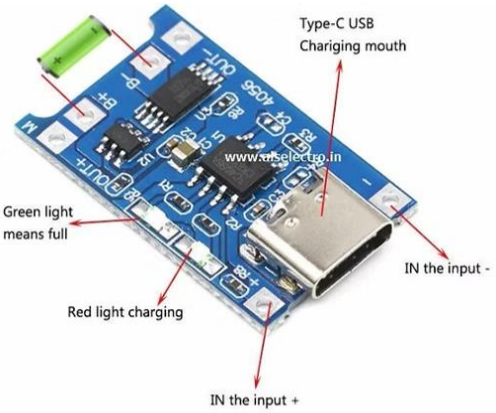


Fig. 3.12. Charger Module

The module used here is TP4056 as its commonly used to charge a 3.7V/4.2V Li-ion batteries.

3.3. SOFTWARE:

1. Arduino IDE: -

The Arduino IDE is an open-source software, which is used to write and upload code to the Arduino boards. The IDE application is suitable for different operating systems such as Windows, Mac OS X, and Linux. It supports the programming languages C and C++. Here, IDE stands for Integrated Development Environment.

It's an open source application to write code, compile it and upload it to a microcontroller, in this, ESP32. It's used as its simple and beginner-friendly interface, built-in code editor with syntax highlighting, easy upload to Arduino boards, large library support and cross-platforms.

In our project, it does:

- Detect the line using IR sensors.
- Follow the line by commanding the Motor Driver and Wheels.
- Detect fire using flame sensors.
- Stop and activate blower to extinguish fire.

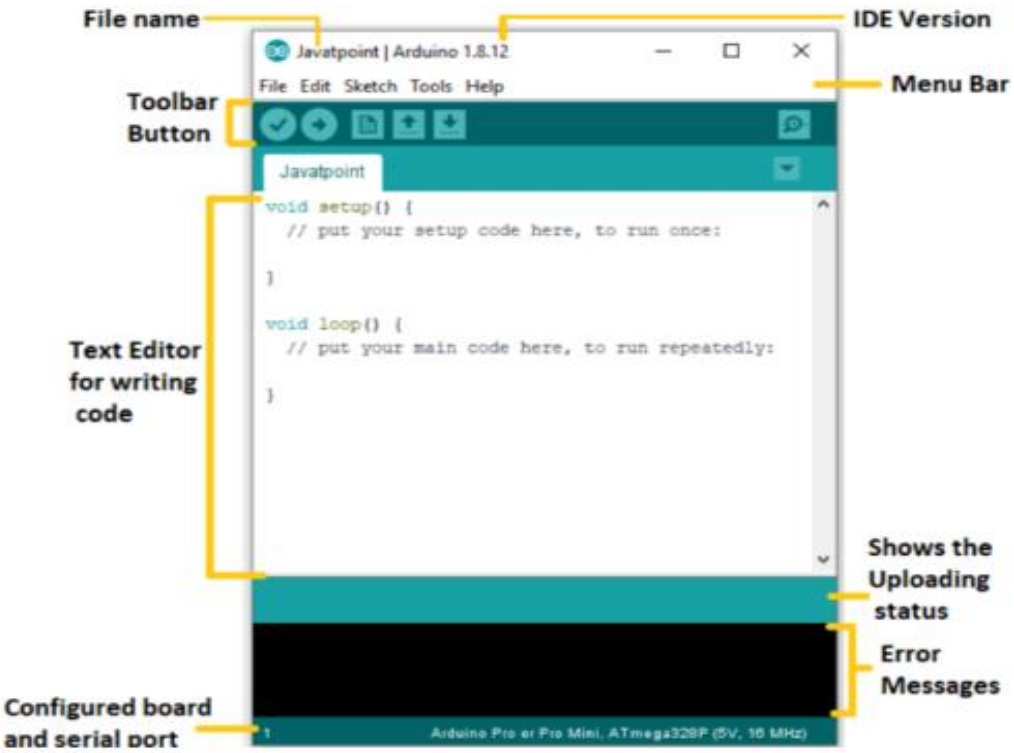


Fig. 3.13. Arduino IDE

4. IMPLEMENTATION

4.1. SYSTEM DESIGN OVERVIEW:

Line follower is an intelligent robot which detects a visual line embedded on the floor and follows it. The path is predefined and can be either visible like a black line on a white surface with a high contrasted colour or the path can be a complex such as magnetic markers or laser guide markers. In order to detect these lines various sensors can be employed. Generally, infrared Sensors are used to detect the line which the robot has to follow. The robot movement is automatic and can be used for long distance application. Line follower can be modified by giving obstacle detection capability to it. If any object is placed on the path, then a normal line follower will try to push the obstacle and hence it gets damaged. By using ultrasonic sensor, the line follower can detect an obstacle and can stop till the obstacle is removed. This type of robots can perform lot of tasks in industries, like material handling. These robots can be used as automated equipment carriers in industries replacing traditional conveyer belts. They also have domestic application and one of the interesting application of this line follower robot is in health care management. As this smart line follower robot has obstacle detection capability it will not be damaged easily as it stops it motion till the obstacle is removed or till the path is changed. This ability of the robot increases it application especially in industries because obstacles are common in any workplace and if the robot is not able to detect the obstruction it will get damaged so this gives an added advantage wherever this intelligent line follower is used.

4.2. HARDWARE INTEGRATION:

i. Controller unit:

- The ESP32 development board acts as the main controller.
- It reads inputs from sensors and controls motors and the blower.

ii. Sensor Module:

- IR Line Sensor Array (5-channel):
  - Detects the black line as path.

- Provides digital signals to determine direction (left, right, straight).

- IR Flame Sensors (3 units):
  - Detects flame presence and direction (left, centre, right).
  - Helps in locating the fire source.

iii. Actuation System:

- DC Gear Motors (3 units):
  - Provide movement to the robot.
  - Controlled using motor driver.
- Motor Driver (L298N):
  - Interfaces motors with ESP32.
  - Controls speed and direction using PWM signals.
- DC Blower (12V)
  - Activated when flame is detected.
  - Used to extinguish fire.

iv. Power Supply:

- Li-ion Battery:
  - Supplies power to ESP32, sensors and motors.
- Buck Converter:
  - Steps down voltage to required levels.

4.3. SOFTWARE DESIGN:

a) *Programming Environment:*

- Code is developed using Arduino IDE and programmed into ESP32.

b) *Control Algorithm:*

i. Initialization:
   - Configure GPIO pins for sensors and motors.
   - Initialize PMW for motor control.
   - Set threshold values for sensors.

ii. Line Following Logic:
   - Continuously read IR sensor array.
   - Based on sensor output:
     - Move forward → if centred
     - Turn left → if line shifts left
     - Turn right → if line shifts right

iii. Flame Detection:
- Continuously monitor flame sensors.
- If flame detected:
  - Stop line following
  - Identify direction of flame

iv. Fire Extinguishing:
- Move towards flame source.
- Activate DC Blower.
- Continue until flame is no longer detected.

v. Resume Operation:
- After extinguishing:
  - Turn off blower
  - Resume line following

4.4. SYSTEM FLOW:

i. Start System
ii. Follow line using IR sensors
iii. Check for flame continuously
iv. If flame detected:
- Stop navigation
- Move towards flame
- Activate blower

v. Extinguish fire
vi. Resume path

4.5. TESTING & CALIBRATION:

i. ***Sensor Calibration:***
- Adjust IR sensitivity for accurate line detection.
- Tune flame sensor thresholds.

ii. ***Motor Calibration:***
- Adjust PMW for balanced movement.

iii. ***System Testing:***

- Test line following independently.
- Test flame detection independently.
- Integrate and test full system.

4.6. FINAL IMPLEMENTATION:

i. Assemble all components on the acrylic chassis.
ii. Ensure proper wiring and secure connections.
iii. Upload final code and perform real-time testing.

5. RESULTS & DISCUSSION

A microcontroller is the brain of the robot, which controls all its functions. It receives input signals from the sensors and generates output signals to the motor driver circuit. The microcontrollers used here is Arduino. A motor driver is an electronic circuit that controls the speed and direction of the motors. Line follower robots usually have 2 DC motors, and the motor driver circuit is responsible for controlling their movements. The most commonly used motor driver is the L293D, which is compatible with many microcontrollers. Line follower robots use various sensors to detect the line and obstacles. The most commonly used sensors are infrared (IR) sensors, which emit and receive IR radiation. The sensors are placed on the underside of the robot, and they detect the contrast between the line and the surface. To detect obstacles, ultrasonic sensors or IR distance sensors can be used.

A power source is required to power the microcontroller, motor driver, and sensors. The power source is a battery which produces a voltage of 9V. The chassis is the physical structure of the robot, which holds all the components together. Here, it's a plastic and its shape and size depend on the requirements. The wheels are attached to DC motors and enable the robot to move forward and backward.

First, attach a motor driver shield to the Arduino. Now connect the motors to the L293D motor driver shield. Motor 1 to motor driver M1, Motor 2 to motor driver 2, connect the IR sensor to motor driver.IR sensor OUT pin is connected to motor driver A0. IR sensor GND pin is connected to motor driver GND pin. IR sensor VCC pin is connected to motor driver 5V pin. Do the same for other IR sensor but make sure that OUT pin is connected to motor driver A1. Connect the Gear motor to motor driver 1 slot. Now doing all connections, it's time to upload the code. Connect the Arduino Uno to PC via USB cable and open the Arduino IDE, select Arduino board, and com port from the tool menu after that upload the given code.

OBSERVATIONS: -

- Line Tracking Performance
- Obstacle Detection Response
- Fire Detection Accuracy
- Blower Effectiveness

5.1. ADVANTAGES: -

a) Autonomous Operation
b) Multi-Functionality
c) Real-Time Response
d) Fast & Predictable Navigation
e) Quick Fire Response

5.2. Limitations: -

i) Strictly Limited to Pre-Defined Paths
ii) Poor Performance in Real-World Fire Scenarios
iii) Sensor Limitations
iv) Limited Fire Suppression
v) Power Consumption
vi) Environmental Sensitivity

5.3. Future scope: -

☑ IoT Integration (Remote Monitoring & Alerts)
☑ Camera-Based Fire Detection
☑ Advanced Fire Extinguishing System
☑ Autonomous Navigation without Line
☑ AI-Based Decision Making
☑ GPS Tracking System
☑ Multi-Robot Coordination
☑ Energy Optimization
☑ Industrial & Smart Building Integration

6. APPLICATIONS

A. Manufacturing and Industrial Plants

- Continuous fire monitoring on production floors
- Quick response to electrical or machine related fires
- Real time alerts to managers through mobile apps

B. Warehouses and Logistics Centres

- Fire detection in storage racks and remote sections
- 24-hour autonomous patrolling
- Immediate suppression in unmanned areas
- Integration with warehouse systems

C. Data Centres

- Fire protection without water damage
- Early detection to protect servers and equipment
- Integration with data centre monitoring systems
- Zone wise monitoring from control room
- Market Potential: Medium to high, growing digital demand

D. Military and Defence Areas

- Autonomous monitoring of base areas
- Fire response for vehicles and equipment
- Reduced human risk in critical zones

E. Oil and Gas Facilities

- Monitoring in hazardous environments
- Fast response to fuel related fires
- Operation in high risk areas
- Instant alerts for abnormal conditions
- Centralized multi-site monitoring

7. CONCLUSION

Robots play a vital role in the global economy and in everyday life. Also of concern for robotic research is the competitiveness and design of patents in the world's industries according to their type of applications. The need for robotic technology is growing in a wide variety of human applications and applications, especially in the manufacturing, medical, service, defence and consumer industries. This linear robot is an example of an industrial robot. By studying this one can create a linear robot for use in industry. Performance can be improved by using good materials and hearing aids. improves motor movement. The cost of setting up a linear robot depends largely on the costly machinery, land, construction and watchmaking equipment to maintain and operate such equipment.

Another way to improve the current system is to replace trained workers with robotic robots. This robot will be able to manage a large number of assets in a manufacturing system in a shorter amount of time, with higher precision and cheaper cost per unit. The applications of the line follower are limited because it cannot be controlled. The only way to control the line follower is to change the path. Using WI-FI module to control the line follower robot will not be helpful because more power will be consumed, so the battery will drain out quickly. Apart from these limitations smart and intelligent line follower robot can be used for long distance applications with a predefined path. This smart and intelligent robot has more benefits because it doesn’t consume much power.

This robotic system can provide an alternative to the existing system by replacing skilled labour, which in turn can perform better tasks with accuracy and lower per capita cost. Line follower robots with obstacle detection have a wide range of potential future works. Here are some possible directions for future research and development Improved obstacle detection: Currently, most line follower robots use simple sensors to detect obstacles, such as infrared sensors or ultrasonic sensors. Future work could explore more advanced sensing technologies, such as LIDAR or computer vision, to 38 improve obstacle detection accuracy and reliability. Autonomous navigation: Most line follower robots require a pre-determined path to follow.

8. REFERENCES

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